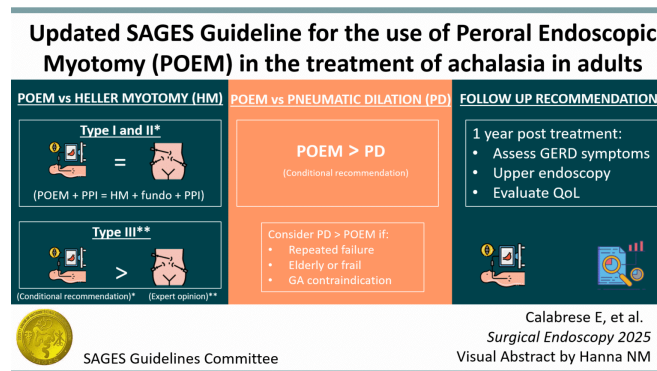




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2024 Update to SAGES Guidelines for the Use of Peroral Endoscopic Myotomy (POEM) in the Treatment of Achalasia

 sages.org/publications/guidelines/update-to-guidelines-for-the-use-of-poem-for-achalasia



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ABSTRACT

Background: Peroral Endoscopic Myotomy (POEM) is a safe treatment for esophageal achalasia with comparable safety to Heller myotomy (HM) plus fundoplication and pneumatic dilatation (PD). More data comparing POEM to Heller Myotomy plus fundoplication and pneumatic dilatation are available since the previous Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) guideline published on this topic in 2021.

Objective: This guideline provides an update to the evidence-based guideline from SAGES published in 2021. It is intended to support clinicians, surgeons, and patients in decisions about the use of POEM for the treatment of achalasia.

Results: The panel agreed on 3 recommendations for adults with achalasia. A conditional recommendation for the use of POEM over PD was issued. Conditional recommendations included the option of using either POEM with appropriate use of proton-pump inhibitors

(PPIs) or HM with fundoplication to treat achalasia. POEM was favored over HM with fundoplication for achalasia subtype III. No new literature was found for children, and the previous recommendation from 2021 should be referred to for this population.

Conclusions: POEM is a safe and effective treatment for achalasia in adults. It demonstrates greater efficacy than PD, though there are specific situations where PD may be the more appropriate choice. POEM has previously been compared to Heller myotomy, but this comparison is less relevant to modern surgical practice. Shifting the paradigm, POEM plus appropriate use of PPIs should be compared to Heller plus fundoplication. This reflection of current practice allows for a conditional recommendation for either approach in the treatment of achalasia subtypes 1 and 2, and for a preference of POEM for subtype 3.

ABBREVIATIONS & ACRONYMS

CI – Confidence Interval

EtD – Evidence to Decision

GERD – Gastroesophageal reflux disease

GRADE – Grading of recommendations, assessment, development, and evaluations

HMF – Heller myotomy with fundoplication

KQ – Key questions

LHM – Laparoscopic Heller Myotomy

PD – Pneumatic dilatation

PICO – Population, intervention, comparison, outcome

POEM – Peroral Endoscopic Myotomy

RCT – Randomized Controlled Trial

RR – Risk ratio

SAGES – Society of American Gastrointestinal and Endoscopic Surgeons

EXECUTIVE SUMMARY

Background

Peroral Endoscopic Myotomy (POEM) was first performed in 2008 and has become widely adopted with over 6,000 cases reported worldwide by 2021 [1]. SAGES published a guideline in 2021 supporting the intervention's safety and efficacy compared to heller myotomy (HM) and pneumatic dilatation (PD) with short term outcomes [2]. In accordance with the SAGES Guideline Update Standard Operating Procedure, published guidelines are reviewed every three years post-publication to reevaluate any new evidence with emphasis on long-term data [3]. This guideline aims to provide updated recommendations for the use of POEM compared to HM and PD in the management of esophageal achalasia.

Interpretation of strong and conditional recommendations

All guideline recommendations were assigned either a “strong” or “conditional” recommendation. The words “the guideline panel recommends” are used for strong recommendations, and “the guideline panel suggests” for conditional recommendations, as per the GRADE approach [4]. A conditional recommendation signals that the benefits of adhering to a recommendation probably outweigh the harms although it does also indicate uncertainty. This uncertainty may be due to a lack of high-quality evidence or variability in how individual patients and other stakeholders value the outcomes of interest.

How to use these guidelines

These guidelines are primarily intended to aid clinicians in making decisions about management of patients diagnosed with achalasia. They are also intended to educate, inform policy and advocacy and to define future research needs. Clinical decision making is multifaceted, and these guidelines are intended to suggest, but not mandate, an acceptable approach to management of achalasia. Finally, these guidelines can also be used by patients as a basis of discussion with their treating surgeon.

Key questions addressed by these guidelines

- 1: Should peroral endoscopic myotomy (POEM) vs. Heller myotomy (HM) be used for achalasia in adults and children?
- 2: Should POEM vs. Pneumatic Dilatation (PD) be used for achalasia in adults and children?

Aim of these guidelines and specific objectives

The purpose of these guidelines is to provide evidence-based, updated recommendations regarding the utility of POEM in the management of esophageal achalasia in adults. These will serve as an update to the guidelines published on this topic in 2021 [2]. It's noted that the update incorporates new data relating to adults. Recommendations for the pediatric population are unchanged as no new comparative data has been identified since the previous guideline. The target audience includes patients, surgeons, gastroenterologists and endoscopists. Policy makers and insurance providers interested in these guidelines include those involved in delivering local, national and international health care services aimed at the treatment of achalasia or to evaluate direct and indirect benefits, harms and costs related to the various procedures used to treat the disease. This document may also serve as the basis for adaptation by local, regional or national guideline panels.

Description of the health problems

Achalasia is an esophageal motility disorder characterized by aperistalsis of the esophageal body and the absence of relaxation of the lower esophageal sphincter (LES) [5]. This results in difficulty with the passage of oral intake from the esophagus into the stomach. It can manifest as symptoms of regurgitation, chest pain, weight loss, and progressive dysphagia.

Other manifestations can be recurrent pneumonia or chronic cough due to aspiration [6]. Achalasia is a relatively rare disorder with a prevalence of about 3 cases in 100,000 and an incidence of 2-3 per 100,000, peaking between the ages of 30 and 60 years of age [7].

The pathophysiology of achalasia is not entirely understood but involves an inflammatory process resulting in loss of the inhibitory ganglion cells in the myenteric plexus of the esophagus and ultimately failure of LES relaxation [6, 8].

Achalasia can also be divided into subtypes on the basis of the pattern of contractility in the esophageal body as measured on the diagnostic test, high-resolution manometry [8]. There are three subtypes which include type I (classical achalasia with no esophageal pressurization), type II (achalasia with esophageal compression) and type III (achalasia with esophageal spasm) [8].

Currently there are no known efficacious pharmacologic treatments that provide long-standing results, and therapy is directed towards manual disruption of the lower esophageal sphincter and decreasing resistance to esophageal emptying. Pneumatic dilatation, Heller myotomy and peroral endoscopic myotomy (POEM) have all been used to lower LES pressure, and each has its benefits and disadvantages. In current practice, Heller myotomy (HM) is nearly always accompanied by fundoplication as an acknowledgement that post-myotomy without fundoplication results in increased rates of gastroesophageal reflux at approximately 30% or higher when objectively diagnosed by either pH testing or by standard endoscopic criteria [9, 10]. These guidelines provide recommendations regarding the use of POEM in the treatment of achalasia.

Methods

The creation of these guidelines followed the SAGES Guidelines Development Standard Operating Procedure [11]. The Guidelines Update Task Force within the SAGES Guidelines Committee performed a systematic review of the literature. The start date for the search was the last date from the previous guideline's literature search. The Cochrane Library, Clinicaltrials.gov, Embase, PubMed, and the International Clinical Trials Registry Platform were searched from 2/18/2019 through 3/8/2023. Search strategies can be found in Appendix A. An important amount of new literature was found, therefore it was decided to move forward with an update of the previous guideline. Of note, the pediatrics literature was reviewed and no new data was found. This guideline updates adult recommendations and the previously published SAGES guideline on POEM for achalasia should be referred to for recommendations regarding children [2].

The systematic review was performed according to the Preferred reporting items for systematic review and meta-analysis (PRISMA) checklist. The new data from this literature search was combined with the data from the previous search to provide a comprehensive

evaluation for each outcome of interest. A new systematic review was not published, as the purpose of this update was to inform the guideline. The PRISMA diagrams can be found in Appendix B.

The leads and panel members from the original project were included when feasible. The guideline panel used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) Evidence to Decision (EtD) approach to deliberate and formulate recommendations [12, 13]. Two patient advocates were also asked for their opinions regarding the outcomes selected and the recommendations voted on by panelists. Reporting of this guideline was structured as per the Essential Reporting Items for Practice Guidelines in Healthcare (RIGHT) checklist [14].

Guideline Panel Organization

International experts in the management of achalasia including POEM who were part of the original guideline panel were invited to participate in the SAGES Guideline Panel. All panel members were experienced endoscopists with training in either surgery or gastroenterology and submitted financial disclosures for review of potential conflicts of interest. An additional voting member was a patient advocate from the International Foundation for Gastrointestinal Disorders (C.R.). A non-voting patient from Achalasia Action (A.W.), and a non-voting patient with no affiliation but with meaningful patient experience (A.J.) were also included in discussion and their input was considered in creating the recommendations. A methodologist with extensive guideline development experience (M.A.) and the SAGES Guidelines Committee Fellow (E.C.) participated in the panel as non-voting members and facilitated appraisal of the evidence and formulation of the recommendations.

The panel reviewed the literature presented by the fellow and met via video conference over the course of two different online sessions to create the recommendation. The panel reviewed evidence tables populated by the systematic review results and voted on components of Evidence to decision tables to reach final recommendations. Both the evidence tables and evidence to decision tables were compiled using GradePro [15]. Evidence tables can be found in Appendix C.

Guideline Funding and Declaration and management of competing interests

Funding for the methodologist, statistician, librarian and guidelines fellow was supported by SAGES Education and Research Foundation (SERF) grant. No industry support was used to create this guideline, nor was any industry input used for any stage of the development, dissemination, or implementation of this guideline. Standard financial disclosure forms from the journal Surgical Endoscopy were completed by all guideline contributors to evaluate for potential conflict of interest. Evaluation was made by the first author, and no potential conflicts were deemed to have affected the recommendations. A list of declarations can be found at the conclusion of this manuscript.

Selection of questions and outcomes of interest

The Guidelines Update Task Force performed a literature search on POEM using the original systematic review search syntax [16]. Heller myotomy and pneumatic dilation were identified as comparators of interest, and key questions were formulated according to the patient-intervention-comparator-outcome (PICO) format. Outcomes that were selected “critical” or “important” for patients were selected *a priori* as outcomes potentially critical or important for decision-making after evidence had been collected. These outcomes centered on efficacy, safety, and side effects associated with POEM, Heller myotomy, and pneumatic dilation. These outcomes included quality of life, dysphagia rates, regurgitation rates, heartburn rates, Eckardt score, proton pump inhibitor (PPI) use, DeMeester score, myotomy length, reintervention due to failure, unexpected leak or perforation, reflux esophagitis, and mortality. Cost was also included. These outcomes were largely based on the previous guideline’s outcomes and then modified based on input from the panel.

Evidence Appraisal

Results from the SAGES systematic review and meta-analysis were uploaded to GradePro to facilitate evidence appraisal and panel decision-making. Evidence that directly compared POEM to either Heller myotomy or PD was used in this guideline. Indirect evidence was not deemed necessary and omitted in favor of direct evidence. Forest plots can be found in Appendix D.

Methods outlined in the GRADE handbook were used to judge the certainty of evidence for each outcome of interest [17]. GRADEPro evidence tables were created. The highest level of data available was used for the tables; less rigorous data that addressed the same outcomes was reviewed but not used in decision making. In brief, the Guideline Fellow and methodologist judged the certainty of the body of evidence across the domains of risk of bias, inconsistency, indirectness, and imprecision. If there was concern in any one of these domains, the certainty was downgraded. This data was then imported into the Evidence to Decision (EtD) table for each KQ. The EtD tables serve as a framework through which the final recommendations are developed. EtD tables can be found in Appendix E.

Assumed Values and Preferences

As this guideline took a patient-centered rather than a societal perspective, the panel members used their collective patient experience to make judgements about patient values and preferences. Two patients and one patient advocate were also consulted directly regarding the outcomes of interest, and their values and preferences surrounding each key question. The two patients were both adult males diagnosed with achalasia. One, from Achalasia Action in England, has been managing their symptoms with diet and PPIs for over a decade. The other underwent POEM in Australia about a year prior to providing input for this guideline. The patient advocate from the International Foundation for Gastrointestinal

Disorders was a North American female, had been involved in the original guideline and has extensive experience speaking with patients that have undergone a wide variety of gastrointestinal procedures. The target audience of this guideline includes patients and clinicians.

Development of clinical recommendations

Outcomes from the evidence table deemed critical and important to decision-making were imported into Evidence to Decision (EtD) tables on GRADEPro. Using the Evidence to Decision tables, the panel discussed desirable effects, undesirable effects, the certainty of evidence, the potential variation in values of key stakeholders, balance of these effects, and acceptability and feasibility of the option favored by the balance of effects. Cost was not considered from a societal perspective, but available cost studies were listed as additional considerations for feasibility and acceptability. After discussing the available evidence for each of these components, as well as pertinent additional considerations noted by the panelists based on interpretation of the evidence or expert experience, the panel would vote on each component of the EtD table. Then, a vote would occur on the final recommendation for that key question. The recommendation was made when $\geq 80\%$ of cast votes were in favor.

The panel addressed subgroups such as achalasia subtypes during a discussion for the justification for their recommendation. These considerations are specified for each key question below. The evidence, additional considerations, and final judgements for each step in the decision-making process are summarized in the recommendations that follow.

Guideline Document review

This guideline was drafted based on the Evidence to Decision tables created by panel voting and discussion. All panel members and working groups members then edited the guideline. The final version of the guideline was then submitted to SAGES Executive board for approval before being published online for public comment for 2 weeks. After this public comment period, the final version of this guideline was submitted for final publication.

KEY QUESTIONS

KQ1: Should peroral endoscopic myotomy with judicious use of PPI versus Heller myotomy (HM) with fundoplication be used for achalasia in adults?

Recommendations:

- The Guideline panel suggests that adult patients with type I and II achalasia may be treated with either POEM with appropriate use of PPI or laparoscopic Heller myotomy with fundoplication based on surgeon and patient shared decision-making. (*conditional recommendation, low certainty of evidence*)
- The panel suggests POEM over laparoscopic Heller myotomy for type III adult achalasia. (*expert opinion*)
- The panel suggests consideration of routine upper endoscopy in both groups to detect the possibility of the asymptomatic patient with significant reflux who could be at risk of adenocarcinoma. The panel also notes an absence of data on this topic and emphasizes the importance of further research to determine exact timing and protocols. The symptomatic patient should undergo appropriate work-up for objective diagnosis of GERD before treatment. (*expert opinion*)

Summary of the evidence

There were a total of 26 studies included in this analysis. Three randomized controlled studies (RCTs) of varying risks of bias depending on the outcome of interest compared POEM versus laparoscopic Heller Myotomy with partial fundoplication (HMF). One of these was included in the previous guideline, and the other two provided new data. Twenty-three observational studies with high risk of bias compared laparoscopic HMF (one study did not specify if laparoscopic or open) with or without partial fundoplication. Twelve of these studies were included in the previous guideline, and eleven studies provided new data. Length of follow-up varied by outcome, but only outcomes with greater than 6-months post-procedure were included in the analysis.

The data below is reported by outcome with the number of studies, total number of participants, certainty of the evidence, statistical method used to evaluate the data with its associated confidence interval (CI), and the absolute risk difference with its associated confidence interval. Odds ratio (OR) was used for dichotomous outcomes and mean difference (MD) and standard mean difference (SMD) for continuous outcomes. The standard mean difference was used rather than mean difference if the rating scales used were not identical between studies. The absolute risk difference for binary outcomes is reported as either more or fewer number of patients/cases occurring in the POEM group relative to the comparator out of 1,000.

Benefits

POEM had similar effectiveness as Laparoscopic Heller myotomy with fundoplication (LHM) with regard to:

- Quality of life (1 RCT with 202 participants, moderate certainty of evidence, standard mean difference 0.18 higher CI [0.09 lower to 0.46 higher])

- DeMeester score, more than 12 months post-procedure (1 RCT with 221 participants, high certainty of evidence, MD 0.8 lower CI [1.97 lower to 0.37 higher])
- Dysphagia, more than 6 months post-procedure (1 RCT with 261 participants, moderate certainty of evidence, OR 0.97 CI [0.49 to 1.91], 4 fewer per 1,000 CI [73 fewer to 104 more])
- Mortality (7 observational studies with 43,418 participants, very low certainty of evidence, OR 0.47 [0.24 to 0.92], 1 fewer per 1,000 CI [1 fewer to 0 fewer])
- Unexpected ICU admission (7 observational studies with 12,071 participants, very low certainty of evidence, OR 0.75 CI [0.19 to 2.98], 0 fewer per 1,000 CI [1 fewer to 1 more])
- Eckardt score (2 RCTs with 80 participants, low certainty of evidence, MD 0.35 higher CI [0.08 lower to 0.78 higher])
- DeMeester score, binary (abnormal being greater than or equal to 14.72 and normal being less than 14.72), greater than 12 months (1 RCT with 125 participants, low certainty of evidence, OR 1.25 CI [0.6 to 2.61], 52 more per 1,000 CI [104 fewer to 233 more])
- Regurgitation at follow up 12 months or longer (1 RCT with 209 participants, moderate certainty of evidence, OR 143 CI [0.24 to 8.41], 8 more per 1,000 CI [15 fewer to 124 more])
- Reintervention for failure (1 RCT with 221 participants, moderate certainty of evidence, OR 1.65 CI [0.38 to 7.08], 17 more per 1,000 CI [7 fewer to 139 more])

POEM was better, to a level determined to be of clinical significance by the panel, in the outcomes of:

Unrecognized leak (2 RCT with 261 participants, very low certainty of evidence, OR 0.66 CI [0.11 to 3.83], 8 fewer per 1,000 CI [21 fewer to 60 more])

The combined magnitude of these effects was determined to be small.

Harms and burden

POEM resulted in more:

- Esophagitis of Grade C or D at greater than 12 months (2 RCTs with 200 participants, very low certainty of evidence, OR 2.18 CI [0.17 to 27.50], 64 more per 1,000 CI [51 fewer to 585 more])
- PPI use greater than 12 months (1 RCT with 209 participants, high certainty of evidence, OR 2.62 CI [1.40 to 4.89], 193 more per 1,000 CI [58 more to 347 more])

The combined magnitude of these effects was determined to be small.

Certainty of evidence

The certainty in these effects was deemed low owing to the risk of bias and imprecision of the estimates.

Decision criteria and additional considerations

Quality of life (QoL) was deemed one of the most critical outcomes by both the panel and the patients. Patients felt that clinical symptoms, such as regurgitation and dysphagia, are directly reflected in quality of life. Our analysis found that quality of life and clinical symptoms of regurgitation and dysphagia were largely equivalent in both POEM patients and HMF patients.

The subjective reporting of heartburn was also reviewed in this update. When attempting to find rates of heartburn following HMF in the literature as indirect evidence of gastroesophageal reflux, the data is limited, as it is typically pooled into the outcome “reflux symptoms” or studies only look at objective tests. Therefore, heartburn was not formally presented due to the lack of evidence for the HMF cohort, making a calculation for the absolute effect compared to POEM unachievable.

Another important consideration is that post-fundoplication side effects are not clearly reported in comparative studies. Late postoperative fundoplication-related side effects can include gas-bloat syndrome and diarrhea, amongst several others [18]. This is an important factor in decision-making and the lack of reporting on these outcomes does not allow for a fair comparison between the two therapies. One patient stated that an important consideration when deciding between POEM and HMF is time away from work and the “size” of the procedure. These were major factors in the patient’s decision to choose POEM. This further highlights the importance of using QoL as a main outcome for success following treatment for achalasia as it encompasses all the above and allows for unreported outcomes to be indirectly considered.

The data show markedly different rates of reflux esophagitis and PPI use reported between the two groups. Objective evidence of severe esophagitis based on Los Angeles Classification (C/D) diagnosed on endoscopy at 12 months or greater post-procedure was used as the measurement of reflux esophagitis. This is one method for objectively evaluating gastroesophageal reflux disease (GERD) and its severity. The treatment goal for management of gastroesophageal reflux is the improvement of symptoms and of quality of life. Symptom control is often achieved with PPI use. While we only used the long-term and most severe forms of esophagitis data to inform this guideline, it is worth noting that rates of esophagitis in the POEM group decreased over time. Pooled reflux rates for esophagitis grades A-D at less than 12 months were greater than over 12 months. This trend persisted with severe grades (C/D) of reflux. It was found that the RCTs evaluating postoperative reflux symptoms included placing patients on a PPI when esophagitis was discovered on follow-up endoscopy, which could explain why rates of esophagitis improved over longer periods of time (see Appendix D for Forest plots) [19–21]. With no other plausible cause identified, it

seems reasonable to assume that POEM plus appropriate PPI use when required has lower rates of esophagitis than POEM alone. The panel considers appropriate use of PPIs to be treating those patients with objectively confirmed abnormal distal esophageal acid exposure.

With this understanding, the discussion shifts to whether asymptomatic esophagitis is relevant, and whether this is the case after a patient is placed on a PPI. The concern is whether asymptomatic esophagitis, or indeed asymptomatic gastroesophageal reflux itself and without esophagitis, leads to increased rates of Barrett's esophagus, esophageal adenocarcinoma, or stricture in the long-term. The long-term effects of severe esophagitis after POEM procedure are not currently reported in the literature. Part of this is because reflux esophagitis can be controlled with PPI use, however, this requires that reflux esophagitis is diagnosed, even in the asymptomatic patient. It is reported that nearly 60% of patients with measured abnormal esophageal acid exposure or evidence of esophagitis during endoscopy have no symptoms [22].

The panel determined that medically uncontrollable reflux, rather than simply reflux esophagitis alone, is a more relevant clinical outcome for both clinicians and patients. Based on experience, the panel discussed anecdotal, rare cases in which patients required a partial fundoplication after POEM for medically uncontrollable reflux, but there is not sufficient evidence on the matter. This phenomenon emphasizes the importance of the need for post-POEM surveillance protocols with objective tests using endoscopy and DeMeester score, to identify these patients and ensure that they are treated or surveilled in a reliable way. The ACGE advises that there is no indication for long term surveillance after POEM because reports of squamous cell carcinoma are low; but this does not take into consideration long-term rates of adenocarcinoma for which there is currently no meaningful data [23, 24].

PPI use as an outcome, while important, is difficult to accurately represent. There is inconsistent reporting due to differences in practice patterns by clinicians, the ability to access this medication over the counter, and because patients may report taking PPIs even if they are not used on a regular basis. Furthermore, prescribing PPIs in patients with achalasia post-procedure is unreliable because symptoms are not a reliable metric of true gastroesophageal reflux. Achalasia symptoms can mimic GERD. Luminal acid, as seen in GERD, and esophageal distention from achalasia can be interpreted by the patient as the same symptom [2]. In addition, several factors need to be considered when interpreting these data, including PPI use for mitigation of severe esophagitis, variability in PPI prescribing, and objective measurement of reflux rather than just symptom-based prescribing. The panel therefore did not believe that the report of PPI use in this data was an accurate representation of GERD and recommended careful history, physical and objective testing before prescribing PPIs.

The panel recognizes that PPI prescribing practices differ between providers after POEM. Some clinicians place patients on PPI directly after POEM or HMF, some base it on subjective symptoms, and others only prescribe using objective data. All of these are

acceptable, however the panel agreed that if no objective testing has been done, it should be performed at some point after prescription, as long-term use of PPIs is not without consequences. The patient advocates were very skeptical of the use of PPIs when asked for their opinion on this outcome and were accepting of short-term use of PPI but not of long-term use. They said long-term use, without specifying an exact timeframe, would weigh into their decision making regarding POEM versus HM. This preference emphasizes the importance of open disclosure with patients about the possible need for PPIs following POEM. It also emphasizes the need for appropriate diagnostics to confirm their need prior to prescription. Further, vulnerable populations may find the use of PPIs limiting due to economic burden or access to healthcare facilities, including a drugstore, e.g. economically vulnerable populations, rural populations, uninsured, or those with limited access to transportation. The panel also discussed subgroup considerations including patients with sigmoid esophagus or Type III achalasia.

These recommendations would seem to permit a paradigm shift away from the previously common practice of simply comparing POEM to Heller myotomy. With the current data, *POEM with appropriate use of PPI* seems to be an efficacious and safe option compared to laparoscopic HM *with partial fundoplication*.

Acceptability and Feasibility

The panel voted that there was possibly important uncertainty or variability particularly due to the outcomes, esophagitis and PPI use. It would be preferred to measure regular PPI use after an objective diagnosis of GERD and measure medically uncontrollable reflux as these are the clinically relevant conditions.

The patient representatives were most concerned with quality of life, symptoms of dysphagia and regurgitation, and long-term PPI use. They stated that our list of outcomes was comprehensive and representative of what patients value.

The panel voted that POEM is feasible to implement relative to HM plus fundoplication. In order to implement POEM, it is necessary for facilities to have adequate infrastructure and clinician training based on society guidelines. This is also required for HMF. Access to diagnostic capabilities preoperatively is critical, such as high-resolution manometry and upper gastrointestinal series. If this is not done properly, patients may receive the wrong intervention, such as Nissen fundoplication for suspected treatment of GERD, or multiple dilations and Botox, especially if manometry is not done. It is also critical that the facilities performing these procedures have the resources to treat possible complications.

The panel also commented that complex procedures should be performed at institutions with high volumes of POEM or HMF cases, such as centers of excellence. Care pathways should be in place for high-volume centers when local hospital resources limit patients' ability to receive standard of care.

Health Equity

Health equity considerations such as third-party payer denials may be a barrier to implementation and should be considered by the clinician and facility performing POEM. Furthermore, those with limited access to healthcare or those that are economically vulnerable may have a more difficult time with PPI access, especially if long-term use. In addition, the learning curve for both POEM and heller myotomy must be considered. Those that do not routinely perform these procedures due to either limited volume or limited facility resources may not achieve equivalent outcomes as those with more experience. Access to facilities that have adequate experience in both procedures may be a challenge for some patients, limiting their decision-making options.

Conclusion

After extensive discussion it was determined that the panel suggests the treatment of achalasia can be either POEM with appropriate use of PPI when required, or alternatively laparoscopic HMF with appropriate use of PPI when required. Quality of Life at 24 months is equivalent after either of these two options.

This is a paradigm shift in that a combination of complementary therapies should be compared rather than only a portion of the widely utilised treatments, that is “POEM + PPI” vs “Heller myotomy + fundoplication” rather than the previous widely compared POEM vs Heller myotomy.

The panel agreed that there is a benefit to POEM in Type III achalasia compared to HMF, but this was based on expert consensus.

Future Research Recommendations

The panel made several recommendations for future research considerations. One of these was a need for studies comparing POEM with appropriate use of PPI to Heller myotomy with fundoplication. In addition, studies with clear indications and monitoring of PPI use and objective measurements of reflux and of post-fundoplication side effects. POEM with PPI use when reflux is diagnosed objectively compared to laparoscopic HM with partial fundoplication would capture many of the gray areas when evaluating these interventions. Comparative studies are needed to determine the rates of subjective heartburn experienced by patients in both POEM and HMF patients. Based on experience, the panel discussed rare anecdotal cases in which patients required a partial fundoplication after POEM for medically uncontrollable reflux, but comparative studies evaluating the frequency of medically uncontrollable reflux with POEM versus HM are needed. Additionally, studies are needed comparing the best way to proceed for failed POEM procedures to determine if this is better approached with HMF or redo POEM.

Volume-based, and experience-based outcomes to determine what is needed from a hospital and clinician perspective for both POEM and HM was also recommended for further study. The panel extensively discussed the importance of surveillance data for optimal follow-up protocols following POEM. Finally, higher quality and larger cost analysis studies comparing POEM and HM need to be performed as this is an important consideration that cannot reliably be considered in decision-making currently.

KQ2: Should peroral endoscopic myotomy vs. Pneumatic Dilatation (PD) be used for achalasia in adults?

Recommendation: The Guideline panel suggests Peroral endoscopic myotomy over pneumatic dilatation (*conditional recommendation, moderate certainty of evidence*).

Summary of the evidence

There were a total of nine studies included in this analysis. Two RCTs, Kuipers et al. and Ponds et al. compared POEM vs PD with high risk of bias when evaluating esophagitis as an outcome due to high drop-out rate, but otherwise low risk of bias for other outcomes [25, 26]. One of these was included in the previous guideline, and the other provided new data. Seven observational studies with high risk of bias compared POEM versus PD. Two of these studies were included in the previous guideline, and five studies provided new data. The length of follow-up varied by outcome, but only outcomes with greater than 6-months post-procedure were included in the analysis. Of note, there was significant selection bias across one of the RCTs and several of the observational studies between the two groups that may have influenced the outcomes regarding achalasia subtype and patient comorbidities, respectively. There was no new comparative evidence for the pediatric patient subgroup.

Benefits

POEM had similar effectiveness as PD with regards to:

- Perforation (7 observational studies with 5,244 participants, very lower certainty of evidence, OR 0.99 CI [0.58 to 1.69], 0 fewer per 1,000 CI [5 fewer to 7 more])
- Unexpected ICU admission (1 observational study with 4,346 participants, very lower certainty of evidence, OR 0.46 CI [0.06 to 3.34], 1 fewer per 1,000 CI [1 fewer to 3 more])

POEM was better to a level determined to be of clinical significance by the panel, in the outcomes of:

- Dysphagia (1 RCT with 125 participants, moderate level of certainty, RR 0.45 CI [0.24 to 0.83], 218 fewer per 1,000 CI [302 fewer to 67 fewer])
- Reintervention for failure (1 RCT with 125 participants, high level of certainty, RR 0.27 CI [0.14 to 0.54], 348 fewer per 1,000 CI [410 fewer to 219 fewer])

The combined magnitude of these effects was determined to be moderate.

Harms and burden

POEM resulted in more:

Esophagitis C/D greater than 12 months (2 RCTs with 141 participants, very low certainty of evidence, OR 2.89 CI [0.34 to 24.82], 52 more per 1,000 CI [20 fewer to 404 more])

The combined magnitude of these effects was determined to be small.

Certainty of evidence

The certainty in these effects is very low owing to the risk of bias and imprecision of the estimates.

Decision criteria and additional considerations

The data were limited regarding outcomes of interest. Of the outcomes recorded in these studies, dysphagia and reintervention for failure rates improved in POEM compared to PD. One patient advocate stated that reintervention is a serious consideration, as most people want long-term symptom reduction. There are select situations in which PD may be preferred to POEM including select elderly and frail patients or in patients with repeated failure after intervention.

Practice patterns related to PD vary by geographic area. The panel discussed that in the United States, this is rarely used in large centers where POEM or HMF are available, however, in Europe it is not uncommon to start with PD for achalasia with serial dilations and then proceed to more definitive management. In their experience, the panel did not find that POEM or HMF was made more difficult in patients with previous PD. There may be a role for PD as a bridge treatment for symptoms in patients awaiting POEM or HMF.

The panel also discussed the importance of distinguishing true anti-achalasia endoscopic dilatation from the more common dilation techniques. For example, distinguishing between through-the-scope (TTS) balloon dilation, which is not a true achalasia dilation procedure, with an achalasia balloon of ≥ 30 mm is important when recommending PD.

Reflux esophagitis rates were reported as higher in the POEM group. As described in KQ1, there are many reasons that this outcome is not an accurate representation of true medically uncontrollable reflux, which is the only clinically relevant outcome related to this.

Furthermore, considering that all other outcomes were determined to be improved with POEM, the panel unanimously felt that POEM is a preferred option to PD when possible.

There are certain subgroup considerations where the use of PD may be considered rather than POEM. These groups include those with repeated treatment failure, the elderly, frail, and patients with the inability to tolerate prolonged procedures and general anesthesia.

Health Equity

It is possible that in certain hospitals there is no infrastructure for POEM nor HMF and PD is the only option for patients in these regions. This also contributes to a health equity consideration, as patients in more rural areas or being cared for in under-resourced hospitals may not be receiving standard of care.

Conclusion

The desirable outcomes for POEM were judged to be moderate in effect size and the undesirable outcomes and undesirable effects for POEM were judged to be small. Thus, the panel suggests the use of POEM over PD, and appropriate use of PPIs when required. The panel agreed that health equity and a patient's comorbidities may play a role in this decision and ultimately both must be taken into consideration when deciding on a treatment. If health inequities present a barrier to accessing definitive management for achalasia, then clear pathways need to be established to ensure patients ultimately receive the standard of care.

Future Research Recommendations

The panel made several recommendations for future research considerations. Among these are comparative studies looking at the role of PD versus POEM after failed HMF or POEM. More comparative studies comparing POEM and PD evaluating QOL as an outcome measure, as well as better objective follow-up data including endoscopic findings, DeMeester scores, and high-resolution manometry. Establishing best practice algorithms for management of persistent dysphagia following dilation including increasing the size of balloon in dilation versus POEM was also discussed as a research priority. Finally, higher quality and larger cost analysis studies comparing POEM and PD need to be performed as this is an important consideration that cannot reliably be considered in decision-making currently.

Implementation of these guidelines

The SAGES Guidelines Committee has several mechanisms for distribution and dissemination of these guidelines for key stakeholders (surgeons, patients). The guidelines will be announced in the SAGES newsletter. *Surgical Endoscopy* is free for all SAGES members and the guidelines are posted on the SAGES website. The guideline is also summarized in a visual abstract which is published on all the above platforms.

Monitoring and evaluation

Factors that should be evaluated for monitoring of this guideline include ensuring that the outcomes are equivalent between both groups, especially long-term. First, reflux esophagitis reported with upper endoscopy more than 1 year postop in both POEM and LHM with fundoplication groups. Similarly, reflux disease with pH testing more than 1 year postop in the symptomatic patient for both POEM and LHM with fundoplication groups. In addition, PPI use as well as uncontrolled reflux despite PPI use more than 1 year postop in both aforementioned groups. Quality of life should be evaluated in both groups, more than 1 year postop. Finally, symptoms of regurgitation, dysphagia more than 1 year postop should be evaluated in both groups.

Updating these guidelines

This is a three-year update of the SAGES POEM Guideline from 2021 [2]. The SAGES Guidelines Update Task Force (GUTF) reevaluates all eligible SAGES guidelines in three years post-publication.[3]

It was agreed that POEM versus PD will be retired. In three years' time, the POEM versus HMF question will be reevaluated by the GUTF and a full update will be decided upon based on predetermined criteria established by the task force protocol.

Limitations of these guidelines

The limitations of these guidelines are inherent to the very low certainty of the evidence for both Key Question 1 and Key Question 2. Furthermore, the data is limited by the short-term follow-up and the lack of outcomes of value in the literature. Outcomes of interest include post-fundoplication side effects, appropriate and reliable PPI use rates, and reliable, objective reflux esophagitis data. The panel expressed an interest in following up on any longer-term data in the future, to abolish the need for this proxy and for more reliable outcomes of interest.

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Declarations

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APPENDICES

[Appendix A – Literature Search Strategies](#)

[Appendix B – PRISMA flow diagram](#)

[Appendix C – Evidence tables](#)

[Appendix D – Forest plots](#)

[Appendix E – Evidence to Decision \(EtD\) tables](#)

References

1. Haisley KR, Swanström LL (2021) The modern age of POEM: The past, present and future of per-oral endoscopic myotomy. *J Gastrointest Surg* 25:551–557.
2. Kohn GP, Dirks RC, Ansari MT, Clay J, Dunst CM, Lundell L Marks JM, Molena D, Rooker C, Saxena P, Swanstrom L, Wong RK, Pryor AD, Stefanidis D (2021) SAGES guidelines for the use of peroral endoscopic myotomy (POEM) for the treatment of achalasia. *Surg Endosc* 35:1931–1948.

3. Calabrese EC, Kumar SS, Kohn GP, Maddern G, Babidge W, Slater BJ et al (2025) Society of American Gastrointestinal and Endoscopic Surgeons guidelines development: clinical practice guideline update standard operating procedure. *Surg Endosc* 39:716–720
4. Andrews JC, Schünemann HJ, Oxman AD, Pottie K, Meerpohl JJ, Coello PA, Rind D, Montori VM, Brito JP, Norris S, Elbarbary M, Post P, Nasser M, Shukla V, Jaeschke R, Brozek J, Djulbegovic B, Guyatt G (2023) GRADE guidelines: 15. Going from evidence to recommendation-determinants of a recommendation's direction and strength. *J Clin Epidemiol* 66:726–735.
5. Pohl D, Tutuian R (2007) Achalasia: an overview of diagnosis and treatment. *J Gastrointest Liver Dis* 16:297–303.
6. Williams VA, Peters JH (2009) Achalasia of the esophagus: a surgical disease. *J Am Coll Surg* 208:151–162.
7. Duffield JA, Hamer PW, Heddle R, Holloway RH, Myers JC, Thompson SK (2017) Incidence of achalasia in South Australia based on esophageal manometry findings. *Clin Gastroenterol Hepatol* 15:360–365.
8. Stavropoulos SN, Friedel D, Modayil R, Parkman HP (2016) Diagnosis and management of esophageal achalasia. *BMJ* 354:i2785.
9. Campos GM, Vittinghoff E, Rabl C, Takata M, Gadenstätter M, Lin F, Ciovica R (2009) Endoscopic and surgical treatments for achalasia: a systematic review and meta-analysis. *Ann Surg* 249:45–57.
10. Richards WO, Torquati A, Holzman MD, Khaitan L, Byrne D, Lutfi R, Sharp KW (2004) Heller myotomy versus Heller myotomy with Dor fundoplication for achalasia: a prospective randomized double-blind clinical trial. *Ann Surg* 240:405–412; discussion 412-415.
11. Rogers AT, Dirks R, Burt HA, Kohn GP, Slater BJ, Walsh D, Stefanidis D, Pryor A (2021) Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) guidelines development: standard operating procedure. *Surg Endosc* 35:2417–2427.
12. Cheyne S, Fraile Navarro D, Hill K, McDonald S, Tunnicliffe D, White H, Whittle S, Karpusheff J, Mustafa R, Morgan RL, Sultan S, Turner T; Australian Living Evidence Consortium Methods and Processes Working Group and Collaborators (2023) Methods for living guidelines: early guidance based on practical experience. Paper 1: Introduction. *J Clin Epidemiol* 155:84–96.
13. Alonso-Coello P, Schünemann HJ, Moberg J, Brignardello-Petersen R, Akl EA, Davoli M, Treweek S, Mustafa RA, Rada G, Rosenbaum S, Morelli A, Guyatt GH, Oxman AD; GRADE Working Group (2016) GRADE Evidence to Decision (EtD) frameworks: a systematic and transparent approach to making well informed healthcare choices. 1: Introduction. *BMJ* 353:i2016.

14. Chen Y, Yang K, Marušić A, Qaseem A, Meerpohl JJ, Flottorp S, Akl EA, Schünemann HJ, Chan ES, Falck-Ytter Y, Ahmed F, Barber S, Chen C, Zhang M, Xu B, Tian J, Song F, Shang H, Tang K, Wang Q, Norris SL; RIGHT (Reporting Items for Practice Guidelines in Healthcare) Working Group (2017) A reporting tool for practice guidelines in health care: The RIGHT statement. *Ann Intern Med* 166:128–132.
15. McMaster University and Evidence Prime. GRADEpro Guideline Development Tool [Software]. 2024. Available: Available from gradepro.org
16. Dirks RC, Kohn GP, Slater B, Whiteside J, Rodriguez NA, Docimo S, Pryor A, Stefanidis D; SAGES guidelines committee (2021) Is peroral endoscopic myotomy (POEM) more effective than pneumatic dilation and Heller myotomy? A systematic review and meta-analysis. *Surg Endosc* 35:1949–1962.
17. Schünemann H, Brožek J, Guyatt G, Oxman A, eds (2013) GRADE handbook. London, UK: Cochrane Collaboration. Available at: <https://gdt.gradepro.org/app/handbook/handbook.html>
18. Richter JE. Gastroesophageal reflux disease treatment: side effects and complications of fundoplication. *Clin Gastroenterol Hepatol*. 2013;11: 465–71; quiz e39.
19. de Moura ETH, Jukemura J, Ribeiro IB, Farias GFA, de Almeida Delgado AA, Coutinho LMA, de Moura DTH, Aissar Sallum RA, Nasi A, Sánchez-Luna SA, Sakai P, de Moura EGH (2022) Peroral endoscopic myotomy vs laparoscopic myotomy and partial fundoplication for esophageal achalasia: A single-center randomized controlled trial. *World J Gastroenterol* 28:4875–4889.
20. Conte TM, Haddad LB de P, Ribeiro IB, de Moura ETH, D’Albuquerque LAC, de Moura EGH (2020) Peroral endoscopic myotomy (POEM) is more cost-effective than laparoscopic Heller myotomy in the short term for achalasia: economic evaluation from a randomized controlled trial. *Endosc Int Open* 8:E1673–E1680.
21. Werner YB, Hakanson B, Martinek J, Repici A, von Rahden BHA, Bredenoord AJ, Bisschops R, Messmann H, Vollberg MC, Noder T, Kersten JF, Mann O, Izbicki J, Pazdro A, Fumagalli U, Rosati R, Germer CT, Schijven MP, Emmermann A, von Renteln D, Fockens P, Boeckxstaens G, Rösch T (2019) Endoscopic or surgical myotomy in patients with idiopathic achalasia. *N Engl J Med* 381:2219–2229.
22. Khashab MA (2020) Reflux after peroral endoscopic myotomy. *Gastroenterol Hepatol (N Y)* 16:417–419.
23. Minami H, Yamaguchi N, Matsushima K, Akazawa Y, Ohnita K, Takeshima F, Nakayama T, Hayashi T, Inoue H, Nakao K, Isomoto H (2013) Improvement of endocytoscopic findings after per oral endoscopic myotomy (POEM) in esophageal achalasia; does POEM reduce the risk of developing esophageal carcinoma? Per oral endoscopic myotomy, endocytoscopy and carcinogenesis. *BMC Gastroenterol* 13:22.
24. Murphy SJ, Anderson LA, Johnston BT, Fitzpatrick DA, Watson PR, Monaghan P, et al. Have patients with esophagitis got an increased risk of adenocarcinoma? Results from a population-based study. *World J Gastroenterol*. 2005;11: 7290–7295.

25. Kuipers T, Ponds FA, Fockens P, Bastiaansen BAJ, Lei A, Oude Nijhuis RAB, Neuhaus H, Beyna T, Kandler J, Frieling T, Chiu PWY, Wu JCY, Wong VWY, Costamagna G, Familiari P, Kahrilas PJ, Pandolfino JE, Smout AJPM, Bredenoord AJ (2022) Peroral endoscopic myotomy versus pneumatic dilation in treatment-naive patients with achalasia: 5-year follow-up of a randomised controlled trial. *Lancet Gastroenterol Hepatol* 7:1103–1111.
26. Ponds FA, Fockens P, Lei A, Neuhaus H, Beyna T, Kandler J, Frieling T, Chiu PWY, Wu JCY, Wong VWY, Costamagna G, Familiari P, Kahrilas PJ, Pandolfino JE, Smout AJPM, Bredenoord AJ (2019) Effect of peroral endoscopic myotomy vs pneumatic dilation on symptom severity and treatment Outcomes among treatment-naive patients with achalasia: A randomized clinical trial: A randomized clinical trial. *JAMA* 322:134–144.

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